

Unit 2 :Linear Software Project Estimation

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2.1 Different Project Estimation Approaches

- **Decomposition techniques**
 - These take a "divide and conquer" approach
 - Cost and effort estimation are performed in a stepwise fashion by breaking down a project into major functions and related software engineering activities
- **Empirical estimation models**
 - Offer a potentially valuable estimation approach if the historical data used to seed the estimate is good .

Software Sizing

- The software size is the most important factor that affects the software cost. This section describes four software size metrics used in practice. The line of code and function point are the most popular metrics among these metrics.

- **Line of Code:**

This is the number of lines of the *delivered source code of the software, excluding* comments and blank lines and is commonly known as *LOC* . Although *LOC* is programming language dependent, it is the most widely used software size metric.

Software Sizing

- **Function points:**
- This is a measurement based on the functionality of the program and was first introduced by Albrecht . The total number of function points depends on the counts of distinct types in the following five classes.
- **Feature point :** *It extends the function points to include algorithms as a new class . An algorithm is defined as the set of rules which must be completely expressed to solve a significant computational problem. This measurement is especially useful for systems with few input/output and high algorithmic complexity, such as mathematical software, discrete simulations, and military applications.*

2.1 COCOMO Model

Constructive Cost Model

- Empirical Model for Software Cost Estimation

COCOMO Technique :

- COCOMO is defined by Boehm in 1981 in his book “Software Engineering Economics”. The COCOMO models are defined for three classes of software projects.
- The software project are classified into three classes using Boehm’s terminology these are :
- *Organic Mode Project*: Relatively small, simple, software projects in which small teams with good application experience work to a set of less rigid requirements.
- *Semi-Detached Mode Project* : An intermediate software project in which teams with mixed experience levels must meet a mix of rigid and less rigid requirements.
- *Embedded Mode Project*: A software project that must be developed within a set of tight hardware, software and operational constraints.

2. effort and development time (TDEV): $TDEV = 2.5 * MM^c$

The coefficients a, b and c depend on the *mode of the development*. There are three modes of development:

Development Mode	Project Characteristics			
	Size	Innovation	Deadline/constraints	Dev. Environment
Organic	Small	Little	Not tight	Stable
Semi-detached	Medium	Medium	Medium	Medium
Embedded	Large	Greater	Tight	Complex hardware/ customer interfaces

Table 1: development modes

1.1 BASIC COCOMO

COCOMO model exist in 3 forms :

Model 1 :Basic COCOMO :

- The basic COCOMO computes software development effort as a function of program size expressed in estimated lines of code.
- It provides an approximate estimation of software cost and is given by following equations :
- $E=A(KLOC)^b$

Where E is effort applied in person-months, KLOC is estimated number of delivered lines of code for project express in thousands, the values of “A” and “b” for different classes of project are as shown in following table :

Basic COCOMO Equation :

Class of Software Project	COCOMO Effort Equation
Organic Mode Project	$E=2.4(KLOC)^{1.05} \text{ pm}$
Semi-detached Mode Project	$E=3.0(KLOC)^{1.12} \text{ pm}$
Embedded Mode Project	$E=3.6(KLOC)^{1.20} \text{ pm}$

Time Calculation :

- Time Required for the development :
- $T_{dev} = B (E)^d$

Class of Software Project	Time Equation
Organic Mode Project	$T = 2.5 (E)^{0.38}$ months
Semi-detached Mode Project	$T = 2.5 (E)^{0.35}$
Embedded Mode Project	$T = 2.5 (E)^{0.32}$

Case study UOP- 2008

- You are required to give cost estimation for the project of size 16,000 LOC . Also Time required for project [10 marks]
- The effort equation of COCOMO as follows:
- $E = A(KLOC)^b$
- As the said project is of Organic mode project. The equation will be substituted with 'A' and 'b' values as follows :
- $$\begin{aligned} E &= 2.4 (KLOC)^{1.05} \\ &= 2.4(16)^{1.05} \\ &= 44 \text{ pm} \end{aligned}$$
$$\begin{aligned} T_{dev} &= 2.5 (E)^{0.38} \\ &= 2.5 (44)^{0.38} \\ &= 10.5 \text{ months} \end{aligned}$$
- *Note : $N = E / T_{dev}$ people*

Case study UOP- May 2009

- You are required to give cost estimation for the project whose size is estimated to be 80,000 LOC . Give the cost estimation for all 3 classes.
- The effort equation of Basic COCOMO as follows:

$$E=A(KLOC)^b$$

- As the said project is of Organic mode project. The equation will be substituted with 'A' and 'b' values as follows :

$$\begin{aligned} E &= 2.4 (KLOC)^{1.05} \\ &= 2.4(80)^{1.05} \\ &= pm \end{aligned}$$

{ Class II : 3.0,1.12 & Class III : 3.6,1.20 }

Dec 2008 :What is use of COCOMO in SPM? (marks 10)

Case Study 2

- The size of an organic type software product has been estimated to be 32,000 lines of source code. Assume that the average salary of software engineers be Rs. 15,000/- per month. Determine the effort required to develop the software product , cost of it and the nominal development time.

Case Study 2

Solution :

From the basic COCOMO estimation formula for organic software:

- The effort equation of COCOMO as follows:
- **$E = A(KLOC)^b$**
- As the said project is of Organic mode project. The equation will be substituted with 'A' and 'b' values as follows :
- $E = 2.4 (KLOC)^{1.05}$
- **Effort = $2.4 \times (32)^{1.05} = 91 \text{ PM}$**
- **Nominal development time = $2.5 \times (91)^{0.38} = 14 \text{ months}$**
- **Cost required to develop the product = $14 \times 15,000$**
= Rs. 210,000/-

Case Study 3

- Suppose that a project was estimated to be 400 KLOC. Calculate the effort and development time for each of the three modes of project i.e., organic, semidetached and embedded.

Solution

- The basic COCOMO equation take the form:

$$E = a (KLOC)^b$$

- Estimated size of the project = 400 KLOC

Case Study 3

(i) Organic mode

$$E = 2.4 (400) 1.05 = 1295.31 \text{ PM}$$

$$D = 2.5(1295.31)0.38 = 38.07 \text{ PM}$$

• **(ii) Semidetached mode**

$$E = 3.0(400)1.12 = 2462.79 \text{ PM}$$

$$D = 2.5(2462.79)0.35 = 38.45 \text{ PM}$$

• **(iii) Embedded mode**

$$E = 3.6(400)1.20 = 4772.81 \text{ PM}$$

$$D = 2.5(4772.8)0.32 = 38 \text{ PM}$$

Case Study 4

- A project size of 200 KLOC is to be developed. Software development team has average experience on similar type of projects. The project schedule is not very tight. Calculate the effort, development time, average staff size and productivity of the project.

- **Solution**
- The semi-detached mode is the most appropriate mode; keeping in view the size, schedule and experience of the development team.

$$E = 3.0 (200)1.12 = 1133.12 \text{ PM}$$

$$D = 2.5(1133.12)0.35 = 29.3 \text{ PM}$$

Hence

- $N = E/D$ persons
- $N = 1133.12/29.3 = 38.67 \text{ Persons}$
- $\text{Productivity} = KLOC / E$
 $= 200 / 1133.12$
 $= 0.1765 \text{ KLOC/PM}$
- $\text{Productivity} = 176 \text{ LOC / PM}$

Software Project Planning

Mode	Project size	Nature of Project	Innovation	Deadline of the project	Development Environment
Organic	Typically 2-50 KLOC	Small size project, experienced developers in the familiar environment. For example, pay roll, inventory projects etc.	Little	Not tight	Familiar & In house
Semi detached	Typically 50-300 KLOC	Medium size project, Medium size team, Average previous experience on similar project. For example: Utility systems like compilers, database systems, editors etc.	Medium	Medium	Medium
Embedded	Typically over 300 KLOC	Large project, Real time systems, Complex interfaces, Very little previous experience. For example: ATMs, Air Traffic Control etc.	Significant	Tight	Complex Hardware/ customer Interfaces required

Table 4: The comparison of three COCOMO modes

Software Engineering (3rd ed.), By K.K Aggarwal & Yogesh Singh, Copyright © New Age International Publishers, 2007



Intermediate COCOMO Model

- The intermediate COCOMO model computes software development effort as function of program size and **a set of “cost drivers”** that include subjective assessment of product, hardware, personnel and project attributes.
- The basic COCOMO model assumes that effort is functions of the product size alone.
- However, a cost of other project parameters besides the product size affect the effort.
- The intermediate COCOMO model recognizes this fact and refines expressions by using a set of 15 Cost Drivers(multipliers) based on various attributes of software development that can be grouped into four major categories .

Categories of Attributes :

- Product attributes
 - Required software reliability
 - Size of application database
 - Complexity of the product
- Hardware attributes
 - Run-time performance constraints
 - Memory constraints
 - Volatility of the virtual machine environment
 - Required time

Categories of Attributes :

- Personnel attributes
 - Analyst capability
 - Software engineering capability
 - Applications experience
 - Virtual machine experience
 - Programming language experience
- Project attributes
 - Use of software tools
 - Application of software engineering methods
 - Required development schedule

Intermediate Cocomo Equation :

- The Intermediate Cocomo formula now takes the form:

$$E = A(KLoC)^b * EAF$$

where E is the effort applied in person-months, KLoC is the estimated number of thousands of delivered lines of code for the project, and EAF is the Effort Adjustment Factor.

- Each of the 15 attributes receives a rating on a six-point scale that ranges from "very low" to "extra high" (in importance or value). An effort multiplier is applied to the rating. The product of all effort multipliers results in an *effort adjustment factor (EAF)*. Typical values for EAF range from 0.9 to 1.4.

Intermediate COCOMO Equation :

Class of Software Project	COCOMO Effort Equation
Organic Mode Project	$E = 3.2(KLOC)^{1.05} * EAF \text{ pm}$
Semi-detached Mode Project	$E = 3.0(KLOC)^{1.12} * EAF \text{ pm}$
Embedded Mode Project	$E = 2.8(KLOC)^{1.20} * EAF \text{ pm}$

EXAMPLE 1 OF INTERMEDIATE COCOMO

- Required: database system for an office automation . Project is organic which has 4 modules to implement:

data entry 0.6 KDSI

data update 0.6 KDSI

query 0.8 KDSI

report generator 1.0 KDSI

- Efforts are rated as follows (all others nominal, 1.0):

cost drivers	level	Ratings
complexity	high	1.15
storage	high	1.06
experience	low	1.13
prog capabilities	low	1.17

Solution:

Note : **System SIZE =3.0 KDSI** ($a=3.2, b = 1.05, d=0.38$)

- $E = A (KLOC)^b * EAF$
- $D = B (E)^d$
- So,
 $E = 3.2 * (3.0)^{1.05} * (1.15 * 1.06 * 1.13 * 1.17 * 1)$
 $E = 3.2 * 3.17 * 1.61$
 $E = 16.33 \text{ PM}$
- $TDEV = 2.5 * (16.33)^{0.38}$
 $TDEV = 7.23 \text{ (>7 months to complete)}$
- How many people should be hired?
$$N = E / TDEV$$
$$= 16.33 / 7.23$$
$$= 2.26$$
$$= 2 \text{ team members}$$

Example: 2

- A new project with estimated 400 KLOC embedded system has to be developed. Project manager has a choice of hiring from two pools of developers: Very highly capable with very little experience in the programming language being used
Or
- Developers of low quality but a lot of experience with the programming language. What is the impact of hiring all developers from one or the other pool ?

You as a Project Manager estimate cost of application.

- **Solution**

This is the case of embedded mode and model is intermediate COCOMO.

Hence,

$$E = A (KLOC)^b * EAF$$

Case I: Developers are very highly capable with very little experience in the programming being used.

$$EAF = 0.82 \times 1.14 = 0.9348$$

$$E = 2.8 (400)^{1.20} * EAF$$

- $E = 3712 \times 0.9348 = 3470 \text{ PM}$
- $D = 2.5 (3470)^{0.32} = 33.9 \text{ M}$

- **Case II: Developers are of low quality but lot of experience with the programming language being used.**
- $EAF = 1.29 \times 0.95 = 1.22$
- $E = 3712 \times 1.22 = 4528 \text{ PM}$
- $D = 2.5 (4528)^{0.32} = 36.9 \text{ M}$
- Case II requires more effort and time. Hence, low quality developers with lot of programming language experience could not match with the performance of very highly capable developers with very little experience.

- **Example 3 :**

- Consider a project to develop a full screen editor. The major components identified are:

I. Screen edit

II. Command Language Interpreter

III. File Input & Output

IV. Cursor Movement

V. Screen Movement

The size of these are estimated to be 4k, 2k, 1k, 2k and 3k delivered source code lines. Use COCOMO to determine

- Overall cost and schedule estimates (assume values for different cost drivers, with at least three of them being different from 1.0)

- **Solution**
- Size of five modules are:
- Screen edit = 4 KLOC
- Command language interpreter = 2 KLOC
- File input and output = 1 KLOC
- Cursor movement = 2 KLOC
- Screen movement = 3 KLOC
- **Total = 12 KLOC**

Let us assume that significant cost drivers are

- i. Required software reliability is high, i.e., 1.15
 - ii. Product complexity is high, i.e., 1.15
 - iii. Analyst capability is high, i.e., 0.86
 - iv. Programming language experience is low, i.e., 1.07
 - v. All other drivers are nominal
-
- $EAF = 1.15 \times 1.15 \times 0.86 \times 1.07 \times 1$
 - $EAF = 1.2169$

- The initial effort estimate for the project is obtained from the
- following equation
- $E = A (KLOC)^b \times EAF$
 $E = 3.2(12)^{1.05} \times 1.2169 = 52.91 \text{ PM}$
- Development time $D = B(E)^d$
 $= 2.5(52.91)^{0.38}$
 $= 11.29 \text{ M}$

Model 3 : Advanced COCOMO

- The advanced COCOMO model incorporates all characteristics of the intermediate version plus an assessment of the cost driver's impact on each step of the software engineering process.

SPPU Case study

Total No. of Questions : 8]

SEAT No. :

P2828

[4775] - 502

[Total No. of Pages : 2

M. C. A. (Management Faculty) (Semester - V)

502 : Software Project Management

(2013 Pattern)

Time : 3 Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) All questions carry equal marks.
- 2) Solve any seven questions from Q. No. 1 to 8.

Q1) Draw a network diagram from the given information. [10]

- a) Find Start Time, End Time, Total Float and Critical Path.
- b) If activity D crashed by 2 weeks, draw network diagram and find out critical path and shortest path.

Activity	ImmediatePredecessor	Duration (in week)
A	-	3
B	A	4
C	-	4
D	C	3
E	D,J	4
F	E,B	3
G	F	3
H	-	5
J	H	3

Q2) A new project with estimated 350 KLOC system has to be developed. For development project also requires. [10]

- a) Software reliability is High (1.15)
- b) Product Complexity is High (1.15)
- c) Analyst Capability is high (0.86)
- d) Programming Language Experience is low (1.07)
- e) Remaining all driver are treated as Nominal.

Calculate the effort, development time, average staff size and productivity of the project.

Total No. of Questions : 7]

P2309

SEAT No. :

[Total No. of Pages : 2

[4675] - 502

M.C.A. (Management Faculty)

IT 52 : SOFTWARE PROJECT MANAGEMENT

(2012 Pattern) (Semester - V)

Time : 3 Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Question 1 & 7 are compulsory.*
- 2) *Solve ANY FOUR Questions from remaining.*
- 3) *Wherever necessary, state Assumptions, give examples & draw diagrams.*
- 4) *Calculators are allowed.*

Q1) a) Draw network diagram from the given information. [10]

- i) Find Start time, End time, Total float & Critical path.
- ii) If activity F is crashed by 2 weeks, draw network diagram & find out shortest path.

Activity	Predecessors	Duration(Weeks)
A	-	2
B	-	3
C	A	4
D	B	2
E	D,C	3
F	B	7
G	B	1
H	E,F	3
I	G	2

b) A project estimated for 251 KLOC has to be developed. for development project also requires: [10]

- i) Software reliability is High (1.15).
 - ii) Product complexity is High (1.15).
 - iii) Analyst capability is high (0.86).
 - iv) Programming language experience is low (1.07).
 - v) Remaining all drivers are treated, as nominal
- calculate effort, Development time period, staff size & productivity.

P.T.O.

SPPU Case Solution with Rubrics

Q.17 b7

③

- Project size = 251 KLOC

- Cost drivers given

{ software reliability is high = 1.15

{ product complexity is high = 1.15

{ Analyst capability is high = (0.86)

{ Programming language experience low = 1.07

{ Remaining cost drivers are minimal = 1.0

i] EAF = Multiplication of Ratings of all cost drives.

$$= 1.15 \times 1.15 \times 0.86 \times 1.07 \times 1.0$$

$$= 1.2169$$

$$\boxed{EAF = 1.2169}$$

2 Marks

ii) Effect - (E)

As size is 251 KLOC, we use Intermediate ^{model} & semi-detached model

$$E = a_i (\text{size})^{b_i} * EAF$$

$$= 3.0 (251)^{1.12} * 1.2169$$

$$\boxed{E = 1778.32 \text{ P-M / M-M}}$$

2 Marks

iii) Development Time period (D)

$$D = c_i (E)^{d_i}$$

$$= 2.5 (1778.32)^{0.35}$$

$$\boxed{CAM = 34.31 \text{ months}}$$

2 Marks

SPPU Case Solution with Rubrics

iv] Staff size (SS)

(4)

$$SS = \frac{E}{D}$$
$$= \frac{1778.32}{34.31}$$

$$SS = 51.83$$

$$SS = 52 \text{ persons}$$

2 Marks

v] Productivity - P

$$P = \frac{\text{size}}{E}$$
$$= \frac{251}{1778.32}$$

$$= 0.1411$$

$$P = 0.1411 \text{ KLOC/P-M}$$

2 Marks

Software Project Planning

Mode	Project size	Nature of Project	Innovation	Deadline of the project	Development Environment
Organic	Typically 2-50 KLOC	Small size project, experienced developers in the familiar environment. For example, pay roll, inventory projects etc.	Little	Not tight	Familiar & In house
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